Q.N.	Questions (ATTEMPT ALL)	Marks	CO	BT
Q.1	What is normalization? Explain 1NF, 2NF & 3NF with example.	5	CO4	BT1
Q.2	Explain different types of integrity constraints.	5	CO3	BT2
Q.3	Consider following schema and represent given statements in relation algebra form.(2 marks each) **Branch**(branch_name, branch_city) **Account** (acc_no, branch_name, balance) **Customer**(customer_id, customer_name, customer_city) **Depositor**(customer_id, acc_no) 1. Find the customer IDs of all customers who live in 'Mumbai' or 'Delhi'. 2. Find the account numbers of accounts with balance between 20,000 and 50,000 3. Find the names of customers along with their account numbers. 4. Find customers who live in 'Pune'. 5. Customer names who live in 'Delhi' but do not have accounts there.	10	CO3	BT3

Q.N.	Questions (ATTEMPT ANY 4)	Marks
Q.1	Draw and explain database system architecture?	5
Q.2	Draw ER model for Library management system and convert it into relational model.	5
Q.3	Write SQL query for following consider table EMP(empno , deptno, ename ,salary, Designation, joining-date, DOB, city) i) Create table for EMP with above scenario. ii) Display names of employees whose experience is more than 10 years. iii) Display age of employees. iv) Display average salary of all employee. v) Display name of employee who earned highest salary.	5
Q.4	Explain fundamental (basic) operations of Relational Algebra.	5
Q.5	Explain the following: 1) Keys in DBMS 2) Aggregate functions in SQL.	5

Q.N.	Question	Marks
Q. 1	Attempt the following	[10]
a	The relationship between EMPLOYEES and EMPLOYER is a a) One-to-one relationship b) One-to-many relationship c) Many-to-many relationship d) Many-to-one relationship	
b	In the database management system, the module which is responsible for choosing efficient execution plan for queries is considered as a) index and glossary optimization b) query processing and optimization c) structural optimization d) content optimization	
c	Internal nodes of the B+-tree form a __________ . a) sparse indices b) multilevel clustered indices c) multilevel dense indices d) multilevel sparse indices	
d	Which SQL statement is used to delete all rows from a table? a) DELETE ALL b) TRUNCATE TABLE c) REMOVE TABLE d) DROP TABLE	
e	SQL's 'ALTER TABLE query' is ______ a) DML b) DDL c) Query Language d) None of these	
f	A relation schema that is in BCNF is also in 3NF. (a) True b) False	
g	The operation on a relation X, produces Y, such that Y contains only a few attributes of X. Such an operation is: a) Projection b) Intersection c) Selection d) Difference	
h	Which of the following is a fundamental concept of a database management system? a) Data structure b) Data integrity c) Data visualization d) Data compression	

i	The_____ operation allows the combining of two relations by merging pairs of tuples, one from each relation, into a single tuple. a) select b) join c) union d) intersection
j	Which of the following is an example of a multivalued attribute in an ER diagram? a) Date of Birth b) Email Address c) Phone Number d) Skills
Q.2	1) Define DBMS? Discuss why is it preferred over file system? 2) Draw an E-R diagram for a hospital
Q.3	1) Explain first normal form with suitable example. 2) Define primary and foreign key relationship between entities with appropriate examples.
Q.4	Explain aggregate functions in SQL with suitable queries (at least three)
Q.5	Explain Armstrong's axioms.
Q.6	1) Define indexed sequential file, primary index and secondary index with example. 2) Differentiate between sparse index and dense index
Q.7	Define a transaction. Draw and explain transaction state diagram.
Q.8	What is a schedule? Explain conflicting instructions? Give an example of a schedule which is conflict serializable.

Max. Marks :60 Marks

Q.N.		Questions (ATTEMPT ANY 6)	Marks	CO	BT
Q.1	a)	Define transaction. Explain briefly different states of transition with neat state transition diagram.	06	CO4	BT2
	b)	Explain ACID properties of transaction.	04	CO1	BT1
Q.2	a)	Explain any 2 locking protocols.	04	CO4	BT2
	b)	Consider the following schema: Suppliers (sid, sname, address) Parts (pid, pname, color) Catalog (sid, pid, cost) Write the relational algebraic queries for the following: i)Find the sids of suppliers who supply some red or green part ii) Find the sids of suppliers who supply every red or green part iii) Find the pids of parts supplied by at least two different suppliers.	06	CO2	BT5
Q.3	a)	Check whether the given schedule is serializable or not? If yes the find the flow of transactions. (schedule below)	06	CO4	BT5
	b)	Explain briefly intrusion detection system.	04	CO4	BT1
Q.4	a)	Given a relation R(P, Q, R, S, T, U, V, W, X, Y) and Functional Dependency set FD = { PQ → R, P → ST, Q → U, U → VW, and S → XY}, determine whether the given R is in 3NF?	05	CO3	BT4
	b)	Explain DDL and DML commands in MySql.	05	CO2	BT3
Q.5	a)	Explain join strategies in detail.	05	CO1	BT4
	b)	Explain access control and its models.	05	CO4	BT1

Schedule for Q.3 a):

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

	b	Draw ER diagram for Banking Management System.	6 M	[illegible]
	c	Write syntax for Create, Alter, Drop and Rename table commands.	6 M	[illegible]
	d	Define Join? Explain different types of joins?	6 M	CO4
	e	Define normalization? Explain 1NF, 2NF, 3NF Normal forms?	6 M	CO5
Q.3		*Solve any four questions. (4x5=20M)*	20 M	
	a	Enlist and explain four aggregate functions.	5 M	CO2
	b	Define Data Abstraction and discuss levels of Abstraction?	5 M	CO3
	c	List any five applications of DBMS.	5 M	CO1
	d	Discuss the Primary and Secondary indexes?	5 M	CO4
	e	Write SQL query for following table EMP(empno , deptno, ename ,salary, Designation, joiningdate, DOB, city) i) Display names of employees whose experience is more than 10 years ii) Display age of employees. iii) Display the average salary of all employees. iv) Display name of employee who earned highest salary.	5 M	CO3
Q.4		*Solve any four questions. (4x5=20 M)*	20 M	
	a	Discuss about different types of Data models?	5 M	CO3
	b	Illustrate different set operations in Relational algebra with an example?	5 M	CO2
	c	Draw and explain database system architecture?	5 M	CO3
	d	Define the following terms i) Key ii) Super key iii) Candidate key iv) Primary key v) Foreign key	5 M	CO2
	e	With an example, explain the concept of View in SQL.	5 M	CO3

	b	Draw ER diagram for Banking Management System.
	c	Write syntax for Create, Alter, Drop and Rename table commands.
	d	Define Join? Explain different types of joins?
	e	Define normalization? Explain 1NF, 2NF, 3NF Normal forms?
Q.3		*Solve any four questions. (4x5=20M)*
	a	Enlist and explain four aggregate functions.
	b	Define Data Abstraction and discuss levels of Abstraction?
	c	List any five applications of DBMS.
	d	Discuss the Primary and Secondary indexes?
	e	Write SQL query for following table EMP(empno , deptno, ename ,salary, Designation, joiningdate, DOB, city) i) Display names of employees whose experience is more than 10 years ii) Display age of employees. iii) Display the average salary of all employees. iv) Display name of employee who earned highest salary.
Q.4		*Solve any four questions. (4x5=20 M)*
	a	Discuss about different types of Data models?
	b	Illustrate different set operations in Relational algebra with an example?
	c	Draw and explain database system architecture?
	d	Define the following terms i) Key ii) Super key iii) Candidate key iv) Primary key v) Foreign key
	e	With an example, explain the concept of View in SQL.

Q.6	a)	Solve by time stamp ordering protocol technique. Provide RTS & WTS table. Whether any transaction will rollback?	05	CO5

T1(ts=100)	**T2**(ts=200)	**T3**(ts=300)
R(A)		
	R(B)	
W(C)		
		R(B)
R(C)		
	W(B)	
W(A)		

	b)	Explain database recovery in detail.	05	CO4
Q.7		Create an employee database using the following fields:	10	CO5

FIELD NAME	DATA TYPE
EMPNO	NUMBER
ENAME	CHAR
DOB	DATE
DEPT	STRING
SALARY	REAL

a) Create the table.
b) Enter 5 tuples.
c) Find sum of salaries of all employees.
d) Find names of employee having highest and least salaries .
e) Delete tuple with second highest salary.

Q. N		Question	Marks	CO	BT
Q.1		*Choose correct option from the following. (5 x 2=10M)*	10 M		
	a	Which of the following is generally used for performing tasks like creating the structure of the relations, deleting relation? 1. DML (Data Manipulation Language) 2. Query 3. Relational Schema 4. DDL (Data Definition Language)	2 M	CO2	BT2
	b	Which of the following provides the ability to query information from the database and insert tuples into, delete tuples from, and modify tuples in the database? 1. DML (Data Manipulation Language) 2. DDL (Data Definition Language) 3. Query 4. Relational Schema	2 M	CO2	BT1
	c	What do you mean by one-to-many relationships? 1. One class may have many teachers 2. One teacher can have many classes 3. Many classes may have many teachers 4. Many teachers may have many classes	2 M	CO2	BT1
	d	The term "FAT" stands for_____. 1. File Allocation Tree 2. File Allocation Table 3. File Allocation Graph 4. All the above	2 M	CO3	BT2
	e	Which one of the following refers to the copies of the same data (or information) occupying the memory space at multiple places. 1. Data Repository 2. Data Inconsistency 3. Data Mining 4. Data Redundancy	2 M	CO3	BT1
2		*Solve following questions. (6x5=30M)*	30 M		
	a	Distinguish any four points between network model and hierarchical model.	6 M	CO3	BT2

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Q.N.	Question	Marks
Q.1	**Attempt any 5 questions.(2 marks each)** **A)** Enlist and explain DDL and DML commands. **B)** State about SELECT and PROJECT operation in Relational algebra? **C)** Discuss the Primary and Secondary indexes? **D)** Define Armstrong axioms for Functional dependency? **E)** List any four applications of DBMS. **F)** Give and state the levels of data abstraction?	**10**
Q.2	**Attempt any 2 questions.(5 marks each)** **A)** Define Join? Explain different types of joins? **B)** Explain about different types of integrity constraints? **C)** Distinguish any four points between network model and hierarchical model.	**10**
Q.3	**Attempt any 2 questions.(5 marks each)** **A)** Write SQL queries for following i) Create table EMP with following attributes using suitable data types (Eno, Ename, Deptname, Salary, designation, Joining_Date) and insert one record. ii) Display names of employee whose name start with alphabet 'A' iii) Display names of employee who joined before '1/1/2000' iv) Increase the salary of employees by 20% who joined after '1/1/2005' .	**10**

C
P

B) Write SQL query for following consider table

EMP(empno , deptno, ename ,salary, Designation, joiningdate, DOB,city)

i) Display employees name and number in an increasing order of salary

ii) Display employee name and employee number dept wise

iii) Display total salary of all employee

iv) Display number of employees dept wise

20

v) Display employee name having experience more than 3 years

vi) Display employee name staring with "S" and working in deptno 1002

C) Consider a schema R (A, B, C, D) and functional dependencies A -> B and C ->D. Solve and find whether the decomposition of R into R1 (A, B) and R2(C, D) belongs to which one or both (dependency preserving and loss less join)?

Join and types of join
Er model and relationship entity attribute and its types
Armstrong axioms
Types of key
Functional dependency and its types
Primary index and secondary index
Sparse index and dark index